

习题 5

1. 求下列函数的傅里叶变换:

$$(1) f(x) = \begin{cases} 1 - x^2, & |x| < 1, \\ 0, & |x| \geq 1; \end{cases}$$

$$(2) f(x) = \begin{cases} e^{-ax} \sin bx, & x \geq 0, \\ 0, & x < 0, \end{cases} \quad \text{其中 } a > 0.$$

2. 求函数 e^{-x} 的傅里叶正弦变换和余弦变换.

3. 求函数 $f(x) = (1 + x^2)^{-1}$ 的傅里叶余弦变换.

4. 设 $a > 0$, b 是任意实数, 证明

$$\int_{-\infty}^{+\infty} e^{-ax^2 - 2bx} dx = \left(\frac{\pi}{a}\right)^{\frac{1}{2}} e^{\frac{b^2}{a}}.$$

5. 设 $F(\alpha) = \mathcal{F}[f(x)]$ 是 $f(x)$ 的傅里叶变换, 证明

$$\mathcal{F}[f(x) \cos \omega x] = \frac{1}{2}[F(\alpha + \omega) + F(\alpha - \omega)];$$

$$\mathcal{F}[f(x) \sin \omega x] = \frac{1}{2i}[F(\alpha - \omega) - F(\alpha + \omega)].$$

6. 已知

$$f_1(x) = \begin{cases} e^{-x}, & x \geq 0, \\ 0, & x < 0; \end{cases} \quad f_2(x) = \begin{cases} \sin x, & 0 < x \leq \frac{\pi}{2}, \\ 0, & \text{其它.} \end{cases}$$

求 $f_1(x) * f_2(x)$.

7. 用傅里叶变换求解下列定解问题:

$$(1) \begin{cases} u_t + au_x = f(x, t), & -\infty < x < +\infty, t > 0, \\ u(x, 0) = \phi(x), & -\infty < x < +\infty; \end{cases}$$

$$(2) \begin{cases} u_{tt} + 2u_t = u_{xx} - u, & -\infty < x < +\infty, t > 0, \\ u(x, 0) = 0, u_t(x, 0) = x, & -\infty < x < +\infty; \end{cases}$$

$$(3) \begin{cases} u_t = a^2 u_{xx}, & x > 0, t > 0, \\ u(x, 0) = f(x), x \geq 0, u_x(0, t) = 0, t \geq 0; \end{cases}$$

$$(4)^* \begin{cases} u_{tt} + a^2 u_{xxxx} = 0, & -\infty < x < +\infty, t > 0, \\ u(x, 0) = f(x), u_t(x, 0) = g''(x), & -\infty < x < +\infty. \end{cases}$$

8. 用正弦变换或余弦变换求解下列定解问题:

$$(1) \begin{cases} u_t = a^2 u_{xx}, & x > 0, t > 0, \\ u(x, 0) = 0, & x \geq 0, u(0, t) = \phi(t), t \geq 0, \\ \lim_{x \rightarrow +\infty} u(x, t) = \lim_{x \rightarrow +\infty} u_x(x, t) = 0; \end{cases}$$

$$(2) \begin{cases} u_{xx} + u_{yy} = 0, & x > 0, y > 0, \\ u(x, 0) = \phi(x), & x \geq 0, \\ u_x(0, y) = 0, & y \geq 0, \\ u(x, y) \text{ 有界.} \end{cases}$$

9. 用二维傅里叶变换方法求解下列定解问题:

$$\begin{cases} u_t = a^2(u_{xx} + u_{yy}), & (x, y) \in \mathbb{R}^2, t > 0, \\ u(x, y, 0) = \phi(x, y), & (x, y) \in \mathbb{R}^2. \end{cases}$$

10. 用傅里叶变换方法求解下列定解问题:

- ✓(1) $u_t = a^2 u_{xx} - tu$ ($x \in \mathbb{R}^1, t > 0$), $u(x, 0) = f(x)$ ($x \in \mathbb{R}^1$);
- (2) $u_t = u_{xx} + cu$ ($x \in \mathbb{R}^1, t > 0$), $u(x, 0) = \delta(x - \xi)$ ($x \in \mathbb{R}^1$);
- (3) $u_t = u_{xx} + 3t^2$ ($x \in \mathbb{R}^1, t > 0$), $u(x, 0) = \sin x$ ($x \in \mathbb{R}^1$);
- (4) $u_t = u_{xx}$ ($x \in \mathbb{R}^1, t > 0$), $u(x, 0) = xe^{-x^2}$ ($x \in \mathbb{R}^1$).